Simple steps to uninstall Git LFS from your repository #3026

Closed **chase-moskal** opened this issue on May 24, 2018 · 21 comments

proven insufficiently formulated, or are simply outdated with deprecated commands (#316, #641, #910)

i was delighted when git lfs was so easy to install and get started with — but it starts to feel frustrating when git lfs is difficult and confusing to break up with — and boy am i confused

so let's make this easy together — and remember — its not you, it's me
ok ok.. i'm seeing someone else.. it's github pages! (github pages doesn't support lfs)

1. **`git lfs uninstall`**
   this removes hooks and smudge/clean filter configuration
   and this is only the beginning, mua-ha-ha

2. ***how can we reindex/re-commit lfs'd files to restore them???***
   in this comment **@andyneff** says:

   > you can probably remove the entries from .gitattributes and trigger the git index that every lfs file is different, and then add and commit those files. This is the same procedure in how you change the auto newlines in a git repo. I'll have to test this some, but this might get you in the right direction. The only caviate is you don't want to reset the .gitattributes files first.

   good thing i read just now that this must be done *before* the .gitattributes step

   but how the heck do we do this? (sure don't want to accidentally change all my newlines, hah)

   is there a way that git lfs can do this for me? after all, it was the one who created this mess to begin with.. *ok, ok, i know that line never works in any relationship ;)*

   edit: ok, there's some talk in this issue about a methodology

   - `git lfs ls-files` — view lfs files
   - for each file, use globs if you can:
     - `git rm --cached myfile.psd` — "remove" the lfs file
     - `git add myfile.psd` — add the "normal" file
   - `git commit -m "restore files from lfs"`

   let me know if that makes sense

3. **remove lfs `.gitattribute` entries**
   this will apparently prevent lfs from resurrecting itself

4. ***~~how do we remove remaining files, seen via `git lfs ls-files`~~ ???***
   i'm guessing we `git lfs untrack` them one-at-a-time
   edit: nope, untrack doesn't remove them... hmm... what does it do?
   maybe once i figure out step 2, this'll be done
   edit: yep, step 2 knocks this one out

5. ***what do we tell our fellow contributors to do???***

also, i have no interest in rewriting git history here, that's silly... i just want to move forward with my collaborators once this is taken care of

oh, and obviously, i need some help with the points ending in lots of question marks — once we work through this, i'd like to edit this post into a nice clean list of steps that people can follow — then maybe let's formalize this into some documentation

thanks fellas

👍 74 😄 6 🎉 2 🚀 13

ttaylorr commented on May 25, 2018 Contributor

Hi **@chase-moskal**, thanks for opening this. I agree with the steps you outlined above, and left a few specific thoughts below. In one sense, I'm glad that we have this issue open in order to point people in the right direction should they want to uninstall Git LFS, but in another sense I think that this 5 (6?) step process is too complicated, and should be handled by Git LFS.

Another confusing aspect of this is that `git lfs uninstall` can be misleading in that it leads users to think that it is "uninstalling" Git LFS from your *repository*, when it is really uninstall Git LFS from your *system*. I think that the ultimate solution to this is a `git lfs migrate export` command.

As you noted, this does not rewrite history, so checking out previous revisions of your repository under Git LFS will check out pointers, not large objects.

Here are some specific thoughts:

> 3. **remove lfs `.gitattribute` entries**
> this will apparently prevent lfs from resurrecting itself

This could occur before step (2) and instead of step(1), which would leave Git LFS installed on your system (in the case that you might interact with a repository that is using it in the future) but would make step (2) do what you expect. Even though Git LFS is configured in `$HOME/.gitconfig`, your repository doesn't have a `.gitattributes` telling Git to invoke Git LFS when you re-run `git add`, so your files are added as expected.

Another caveat of this case is that you must have a local cache of all Git LFS objects for this to work, which you can accomplish by `git lfs pull --all`.

> 5.
> ***what do we tell our fellow contributors to do???***
> i'm guessing they just need to `git lfs uninstall` ?
> do they also need to `git lfs untrack` stored files or anything like that?

They should simply pull and checkout the new state of the world, their repository will work as-expected without Git LFS installed.

👍 2 😄 1

ttaylorr closed this as completed on May 25, 2018

WtfJoke mentioned this issue on May 28, 2018

Undo git lfs migrate import #3032

Open

fedor57 commented on Jan 4, 2019

Thanx, that helped me a lot while I was migrating from BitBucket to AWS CodeCommit. Some helpful hints:

- commit & push everything
- remove hooks
  `git lfs uninstall`
- remove lfs stuff from `.gitattributes`
- list lfs files using
  `git lfs ls-files | sed -r 's/^.{13}//' > files.txt`
- run git rm --cached for each file
  `while read line; do git rm --cached "$line"; done < files.txt`
- run git add for each file
  `while read line; do git add "$line"; done < files.txt`
- commit everything
  `git add .gitattributes`
  `git commit -m "unlfs"`
  `git push origin`
- check that no lfs files left
  `git lfs ls-files`
- remove junk
  `rm -rf .git/lfs`
- you're all done
  (but unlinked junk is within BitBucket Git LFS storage still)

👍 194 👎 1 😄 15 🎉 27 ❤️ 45 🚀 32 👀 7

tchebb commented on Jan 9, 2019

As of Git 2.16, the gnarly piece of this process—re-staging the files that were previously in LFS so their contents are in Git again—can be handled entirely using `git add --renormalize .`. See my Stack Overflow answer here for more details.

ttaylorr commented on Jan 25, 2019 Contributor

> can be handled entirely using `git add --renormalize .`

To clarify, this does *not* affect earlier revisions of history: if you "remove" Git LFS from your repository and check out a historical revision of your repository, you will not see the large files, and instead only see the pointer representation.

`git lfs migrate export` *can* do this, and more information on that is available via `man git-lfs-migrate`.

👍 6 😄 1

ttaylorr mentioned this issue on Feb 13, 2019

How to remove LFS completely in my repository? #910

Closed

crvernon mentioned this issue on Jul 17, 2019

Remove GitLFS Dependency JGCRI/xanthos#40

Open

shasheene added a commit to shasheene/rescuezilla.github.io that referenced this issue on Oct 27, 2019

`Removes Git LFS because no GitHub Pages support` ··· `ea2e42f`

altschuler commented on Nov 22, 2019

LPT: do not remove `.git/lfs` before you push, or you'll be stuck with lfs failing to push

mjh316 mentioned this issue on Apr 17, 2020

Uploading LFS objects: 0% (0/1), 6.5 GB Object increases in size again and doesn't upload #4105

Closed

perumaal mentioned this issue on May 15, 2020

git lfs uninstall does not work with windows #2416

adarshmathew mentioned this issue on Jun 4, 2020

Enable Git LFS support bakerwho/weboftruth#3

Closed

sacdallago pushed a commit to sacdallago/bio_embeddings that referenced this issue on Jul 9, 2020

`Remove git lfs` ··· `8a30567`

herschel666 mentioned this issue on Jul 20, 2020

Remove git-lfs 🤷‍♂️ herschel666/photoblog#483

Closed

everttrollip commented on Aug 11, 2020

I summarised everything mentioned above into a gist, please check it out, maybe it helps others in the future like it helped me. Thanks everyone! https://gist.github.com/everttrollip/198ed9a09bba45d2663ccac99e662201

❤️ 3

long-nguyenxuan commented on Aug 31, 2020

Especially thanks for your script, it saved ton of my times. **@fedor57**
After do all your steps, I've get rid of LFS in my current repository and current remote,
But because I need to push my repository to new remote (gitlab), then when I run push with `git push new-remote --all` , it throw error, Can you give a look on it?

```
longnx@DESKTOP-D46DU6R MINGW64 /d/git/working/edp-ibs-jidou (master)
$ git push gitlab
Enumerating objects: 444, done.
Counting objects: 100% (444/444), done.
Delta compression using up to 4 threads
Compressing objects: 100% (245/245), done.
Writing objects: 100% (444/444), 19.27 MiB | 4.65 MiB/s, done.
Total 444 (delta 190), reused 377 (delta 182), pack-reused 0
remote: Resolving deltas: 100% (190/190), done.
remote: GitLab: LFS objects are missing. Ensure LFS is properly set up or try a manual
To https://gitlab.com/dps-ss/edp-ibs-jidou.git
 ! [remote rejected] master -> master (pre-receive hook declined)
error: failed to push some refs to 'https://gitlab.com/dps-ss/edp-ibs-jidou.git'
```

easiest way to do it that worked for me :
first commit & push everything
then

```
git lfs uninstall
```

then manually remove lfs stuff from .gitattributes

```
git lfs untrack '<pattern>' (in my case, all files "*.*")
git add --renormalize .
git commit -m 'Restore file contents that were previously in LFS' ```
```

👍 23 ❤️ 9

fjuniorr added a commit to fjuniorr/remuneracao-fjuniorr that referenced this issue on Dec 2, 2020

`Run git add --renormalize . from git-lfs/git-lfs#3026` `20730f3`

bjennings76 commented on Jan 17, 2021 • edited

> Especially thanks for your script, it saved ton of my times. **@fedor57**
> After do all your steps, I've get rid of LFS in my current repository and current remote,
> But because I need to push my repository to new remote (gitlab), then when I run push with `git push new-remote --all` , it throw error, Can you give a look on it?
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> ```
> longnx@DESKTOP-D46DU6R MINGW64 /d/git/working/edp-ibs-jidou (master)
> $ git push gitlab
> Enumerating objects: 444, done.
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> Total 444 (delta 190), reused 377 (delta 182), pack-reused 0
> remote: Resolving deltas: 100% (190/190), done.
> remote: GitLab: LFS objects are missing. Ensure LFS is properly set up or try a
> manual "git lfs push --all".
> To https://gitlab.com/dps-ss/edp-ibs-jidou.git
> ! [remote rejected] master -> master (pre-receive hook declined)
> error: failed to push some refs to 'https://gitlab.com/dps-ss/edp-ibs-
> jidou.git'
> ```

FYI, I had this same error when I tried to push a repo to GitLab that at one time had LFS files in its

2. During the rebase changed the .gitattribute file to remove all references to the LFS file type
3. Removed/squashed commits that originally changed the LFS files I had swapped out when removing LFS via commits
4. Continued the rebase
5. Force-pushed the new commits up to the upstream repo

After that, I was able to push the repo up to GitLab without the `LFS objects are missing` error.

gasilva commented on Jan 27, 2021

> easiest way to do it that worked for me :
> first commit & push everything
> then
>
> ```
> git lfs uninstall
> ```
>
> then manually remove lfs stuff from .gitattributes
>
> ```
> git lfs untrack '<pattern>' (in my case, all files "*.*")
> git add --renormalize .
> git commit -m 'Restore file contents that were previously in LFS' ```
> ```

That worked well!!! Thank you.

MrOrz mentioned this issue on Feb 11, 2021

Remove git lfs & files cofacts/opendata#20

Merged

garrettheath4 added a commit to garrettheath4/zines that referenced this issue on Feb 15, 2021

`Patch: Disabled and removed Git LFS due to incompatibility issues wit…` `fa3b92b`

...

pratik3098 commented on Feb 20, 2021

Removing the following files worked for me:
.git/lfs
.git/hooks/pre-push

megahirt mentioned this issue on Mar 12, 2021

remove git lfs mseag/bible-karaoke#142

Merged

lsetiawan mentioned this issue on May 5, 2021

Are we fully deprecating use of Git LFS? OSOceanAcoustics/echopype#271

Closed

hugomarquesdev commented on Jun 1, 2021

> easiest way to do it that worked for me :
> first commit & push everything
> then
>
> ```
> git lfs uninstall
> ```
>
> then manually remove lfs stuff from .gitattributes
>
> ```
> git lfs untrack '<pattern>' (in my case, all files "*.*")
> git add --renormalize .
> git commit -m 'Restore file contents that were previously in LFS' ```
>
> ```

So simple and it worked!! Thanks 😄

😄 1

gasilva commented on Jun 1, 2021 via email

The git add did not work.

Nothing specified, nothing added.

hint: Maybe you wanted to say 'git add .'?

hint: Turn this message off by running

hint: "git config advice.addEmptyPathspec false"

Em ter., 1 de jun. de 2021 às 06:41, Hugo Marques ***@***.***>
escreveu:
...

gasilva commented on Jun 1, 2021 via email

It was missing a dot after renormalize

git add --renormalize .

Now it worked

Guilherme A. Lima da Silva

Em ter., 1 de jun. de 2021 às 10:35, Guilherme Araujo Lima da Silva <
@.***> escreveu:
...

gasilva commented on Jun 1, 2021 via email

But LFS it is still there

remote: Resolving deltas: 100% (90/90), completed with 46 local objects.

remote: *warning*: GH001: Large files detected. You may want to try Git
Large File Storage - https://git-lfs.github.com
<https://signaldmain.website/click?redirect=https%3A%2F%2Fgit-lfs.github.com&
dID=1622556604166&linkName=https://git-lfs.github.com>
.

remote: *warning*: See http://git.io/iEPt8g
<https://signaldmain.website/click?redirect=http%3A%2F%2Fgit.io%2FiEPt8g&dID=1622556604166&
linkName=http://git.io/iEPt8g>
for more information.

remote: *warning*: File countries/historyIC_v2.ipynb is 89.84 MB; this is
larger than GitHub's recommended maximum file size of 50.00 MB

remote: *error*: GH008: Your push referenced at least 10 unknown Git LFS
objects:

remote: 25b74c60ba87ffc7e6d24da2ecffbc98e90833e890742d5d06f7d7fa8fbc1b96

remote: 4a6428cb65e7a4035753589d2f3af12f6441ba89b3712134006b0cb38134b983

remote: ...

remote: Try to push them with 'git lfs push --all'.

To https://github.com/gasilva/dataAndModelsCovid19.git
<https://signaldmain.website/click?redirect=https%3A%2F%2Fgithub.com%2Fgasilva%2FdataAndModelsCovid19.git&dID=1622556604166&linkName=https://github.com/gasilva/dataAndModelsCovid19.git>

Guilherme A. Lima da Silva

Em ter., 1 de jun. de 2021 às 11:07, Guilherme Araujo Lima da Silva <
@.***> escreveu:
...

andyneff commented on Jun 2, 2021 Contributor

@gasilva That error message is an error from the "GitHub server", not from you running the "git-lfs client".

That error message says you have a file the violated the GitHub max:
`countries/historyIC_v2.ipynb`

GitHub has a policy, where they no longer allow you to push a commit that contains files larger than 50MB. So the "GitHub server" is telling you "You need git lfs"

EvanDorsky added a commit to EvanDorsky/evandorsky.github.io that referenced this issue on Jul 9, 2021

`Remove lfs files, github pages doesn't support lfs` ··· `7cd65a8`

devBabinski commented on Jul 18, 2021

This worked for me:

git lfs uninstall
git lfs uninstall --local
git rm --cached -r .
git push origin main

longtth commented on Jul 20, 2021

> This worked for me:
>
> git lfs uninstall
> git lfs uninstall --local
> git rm --cached -r .
> git push origin main

did you tried clone, update, commit, push from another local repo?
I've tried this and it still stuck

👍 1

JoerivanEngelen mentioned this issue on Jan 10, 2022

Revert icons via Git LFS ASAP Deltares/imod-qgis#22

Closed

ann0see mentioned this issue on Jan 26, 2022

Use LFS to save space jamulussoftware/jamuluswebsite#556

Closed

2 tasks

florianseidlschulz commented on Apr 4, 2022

> Thanx, that helped me a lot while I was migrating from BitBucket to AWS CodeCommit. Some helpful hints:
>
> ```
> * commit & push everything
>
> * remove hooks
>  `git lfs uninstall`
>
> * remove lfs stuff from `.gitattributes`
>
> * list lfs files using
>  `git lfs ls-files | sed -r 's/^.{13}//' > files.txt`
>
> * run git rm --cached for each file
>  `while read line; do git rm --cached "$line"; done < files.txt`
>
> * run git add for each file
>  `while read line; do git add "$line"; done < files.txt`
> ```

```
    `git commit -m "unlfs"`
    `git push origin`

  * check that no lfs files left
    `git lfs ls-files`

  * remove junk
    `rm -rf .git/lfs`

  * you're all done
    (but unlinked junk is within BitBucket Git LFS storage still)
```

This does not work, git lfs ls-files still produces a list of tracked files. Also misses a rm files.txt

👍 2

yjlolo added a commit to yjlolo/dmelodies_dataset that referenced this issue on May 12, 2022

`follow git-lfs/git-lfs#3026 (comment) to unlfs` `8af3642`

Aiyuative commented on Jul 15, 2022 • edited

I just ended up git rm "all the tracked git lfs tracked files".

I dont want to deal with any git rewriting and this allows me to continue my work:)

bendnorman mentioned this issue on Jul 20, 2022

Git LFS bandwidth issue catalyst-cooperative/pudl-usage-metrics#60

Open

Coriago mentioned this issue on Feb 18

Fix could not read username error in terminal RoboEagles4828/edna2023#10

Closed

PasaOpasen commented on Apr 24 • edited

I used

```
rm -rf .git/lfs .git/hooks/pre-push .gitattributes
```

Jython1415 mentioned this issue last week

BUG/DEV: Remove the dependency on Git LFS Jython1415/zoll-foundation-grant-project#74

Open

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

No branches or pull requests

18 participants